

Problem

You are given a directed flow network consisting of V vertices and E directed edges. Each edge has a capacity that represents the maximum amount of flow that can pass through it.

The source vertex is 0 and the sink vertex is $V-1$.

Your task is to find the maximum possible flow from the source to the sink using the Ford—Fulkerson algorithm.

Submissions

Input Format

The first line contains two space-separated integers: V — number of vertices E — number of directed edges Each of the next E lines contains three space-separated integers: u — starting vertex v — ending vertex capacity — capacity of the edge

Leaderboard

Constraints

$2 \leq V \leq 100$ $1 \leq E \leq 1000$ $0 \leq u, v < V$ $u \neq v$ $1 \leq \text{capacity} \leq 10^6$

Output Format

Print a single integer representing the maximum flow from vertex 0 to vertex $V-1$.

Discussions

Sample Input 0

```
6 9
0 1 16
0 2 13
1 2 10
2 1 4
1 3 12
3 2 9
2 4 14
4 3 7
3 5 20
```

Sample Output 0

```
19
```

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Language Python 3



```
1 from collections import deque
2 def ford_fulkerson(g, n):
3     flow = 0
4     while True:
5         parent = [-1] * n
6         parent[0] = 0
7         q = deque([0])
8         while q:
9             u = q.popleft()
10            for v in range(n):
11                if parent[v] == -1 and g[u][v]:
12                    parent[v] = u
13                    q.append(v)
14            if parent[n - 1] == -1:
15                break
16            f = float('inf')
17            v = n - 1
18            while v != 0:
19                u = parent[v]
20                f = min(f, g[u][v])
21                v = u
22            v = n - 1
23            while v != 0:
```

Line: 36 Col: 1



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0

Compiler Message

Success